Experiment No 2a

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Aim: Tutorial on Solving exercises in Data Exploration.

Objective:
After completing this experiment, you will be able to understand the exploratory Data Analysis.

LO's to be achieved:

LON	Lab Outcomes
LO2	Identify sources of Data for mining and perform data exploration

Theory:

Having in mind that data mining is an analytic process designed to explore large amounts of data in search of consistent and valuable hidden knowledge, the first step made in this fabulous research field consists in an initial data exploration.

The exploratory data analysis makes use of the human ability to recognize patterns based on the previous experience. Based on information and knowledge accumulated over time, people can recognize certain forms, trends, patterns, etc., systematically appearing in data, and that cannot always be emphasized by classical methods of investigation. Thus, to choose the optimal data mining methodology for the available data, we need a data analysis, an exploration of them with well-known statistical means.
Basically, exploratory data analysis (EDA) is the Statistics part which deals with reviewing, communicating and using data in case of a low level of information on them. EDA uses various techniques -many of them based on visualization- in order to:

- Maximize the innermost knowledge of the data;
- Reveal underlying structure;
- Extract important variables;
- Detect outliers/anomalies;
- Identify fundamental assumptions to be tested afterwards;
- Develop simple enough models;
- Determine the optimal setting of parameters;
- Suggest some hypotheses concerning the causes of the observed phenomena;
- Suggest appropriate statistical techniques for the available data;
- Provide knowledge for further data collection in support of research or experimentation.

EDA techniques are used as a preamble to the data mining process; the results must be verified and validated by applying it to different data sets to tests its quality.

Basic Statistical Descriptions of Data

For data preprocessing to be successful, it is essential to have an overall picture of your data. Basic statistical descriptions can be used to identify properties of the data and highlight which data values should be treated as noise or outliers.

The three areas of basic statistical descriptions are:

- **Measures of central tendency**-- which measure the location of the middle or center of a data distribution. Intuitively speaking, given an attribute, where do most of its values fall? In particular, we discuss the mean, median, mode, and midrange.
- **Dispersion of the data** -- how are the data spread out? The most common data dispersion measures are the range, quartiles, and interquartile range; the five-number summary and boxplots; and the variance and standard deviation of the data. These measures are useful for identifying outliers.
- **The Graphic displays** of basic statistical descriptions to visually inspect our data. Most statistical or graphical data presentation software packages include bar charts, pie charts, and line graphs. Other popular displays of data summaries and distributions include quantile plots, quantile–quantile plots, histograms, and scatter plots.

: Measuring the Central Tendency: Mean, Median, and Mode

If we were to plot the observations for income, where would most of the values fall? This gives us an idea of the central tendency of the data. Measures of central tendency include the mean, median, mode. The most common and effective numeric measure of the "center" of a set of data is the (arithmetic) mean. The **mean** of set of values is

$$\bar{x} = \frac{\sum_{i=1}^{N} x_i}{N} = \frac{x_1 + x_2 + \cdots + x_N}{N}.$$

This corresponds to the built-in aggregate function, *average* (avg() in SQL), provided in relational database systems. Although the mean is the single most useful quantity for describing a data set, it is not always the best way of measuring the center of the data. A major problem with the mean is its sensitivity to extreme (e.g., outlier) values. Even a small number of extreme values can corrupt the mean. For example, the mean salary at a company may be substantially pushed up by that of a few highly paid managers.

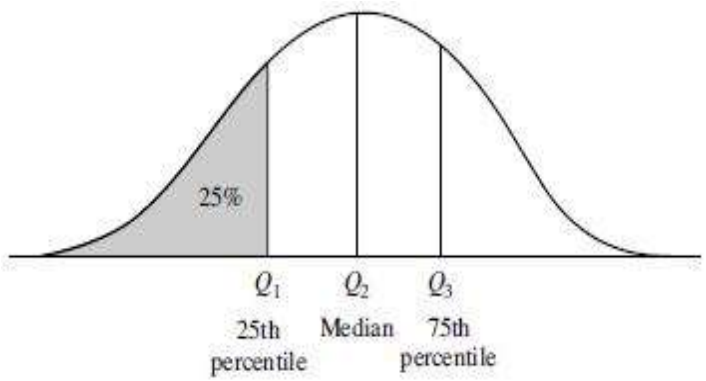
For skewed (asymmetric) data, a better measure of the center of data is the **median**, which is the middle value in a set of ordered data values. It is the value that separates the higher half of a data set from the lower half.

The *mode* is another measure of central tendency. The **mode** for a set of data is the value that occurs most frequently in the set. Therefore, it can be determined for qualitative and quantitative attributes. It is possible for the greatest frequency to correspond to several different values, which results in more than one mode. Data sets with one, two, or three modes are respectively called unimodal, bimodal, and trimodal. In general, a data set with two or more modes is multimodal. At the other extreme, if each data value occurs only once, then there is no mode.

Measuring the Dispersion of Data: Range, Quartiles, Variance &Standard Deviation

The dispersion or spread of numeric data includes range, quantiles, quartiles, percentiles, and the interquartile range. The five-number summary, which can be displayed as a boxplot, is useful in identifying outliers. Variance and standard deviation also indicate the spread of a data distribution.

The range of the set is the difference between the largest (max()) and smallest (min()) values. Suppose that the data for attribute X are sorted in increasing numeric order. Imagine that we can pick certain data points so as to split the data distribution into equal-size consecutive sets, as in Figure-1. These data points are called quantiles. Quantiles are points taken at regular intervals of a data distribution, dividing it into essentially equalsize consecutive sets.



- Figure-1 A plot of the data distribution for some attribute *X*. The quantiles plotted are quartiles. The three quartiles divide the distribution into four equal-size consecutive subsets. The second quartile corresponds to the median.

The 2-quantile is the data point dividing the lower and upper halves of the data distribution. It corresponds to the median. The 4-quantiles are the three data points that split the data distribution into four equal parts; each part represents one-fourth of the data distribution. They are more commonly referred to as quartiles. The 100-quantiles are more commonly referred to as percentiles; they divide the data distribution into 100 equal-sized consecutive sets. The median, quartiles, and percentiles are the most widely used forms of quantiles.

The quartiles give an indication of a distribution's center, spread, and shape. The distance between the first and third quartiles is a simple measure of spread that gives the range covered by the middle half of the data. This distance is called the interquartile range (IQR) and is defined as

$$IQR = Q_3 - Q_1$$

No single numeric measure of spread (e.g., *IQR*) is very useful for describing skewed distributions.

Therefore, it is more informative to also provide the two quartiles $Q1$ and $Q3$, along with the median. A common rule of thumb for identifying suspected outliers is to single out values falling at least 1.5 x *IQR* above the third quartile or below the first quartile.

Because $Q1$, the median, and $Q3$ together contain no information about the endpoints (e.g., tails) of the data, a fuller summary of the shape of a distribution can be obtained by providing the lowest and highest data values as well. This is known as the ***five-number summary***. The five-number

summary of a distribution consists of the median ($Q2$), the quartiles $Q1$ and $Q3$, and the smallest and largest individual observations, written in the order of *Minimum*, $Q1$, *Median*, $Q3$, *Maximum*.

Example of Five-number summary

- For numerical attribute values
- (minimum, Q1, Q2, Q3, maximum)
- Attribute values: 6 47 49 15 42 41 7 39 43 40 36
- Sorted: 6 7 **15** 36 39 **40** 41 42 **43** 47 49
- Q1 = 15 lower quartile
- Q2 = median = 40 (mean = 33.18)
- Q3 = 43 upper quartile
- Q3 – Q1 = 28 interquartile range

Variance and standard deviation are measures of data dispersion. They indicate how spread out a data distribution is. A low standard deviation means that the data observations tend to be very close to the mean, while a high standard deviation indicates that the data are spread out over a large range of values.

The **variance** of N observations, $x_1, x_2, \ldots, x_N$, for a numeric attribute X is

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

Where $\bar{x}$ is the mean value of the observations. The standard deviation, σ, of the observations is the square root of the variance, σ^2.

Histograms

Plotting histograms is a graphical method for summarizing the distribution of a given attribute, X. If X is nominal, such as *automobile model* or *item type*, then a pole or vertical bar is drawn for each known value of X. The height of the bar indicates the frequency (i.e., count) of that X value. The resulting graph is more commonly known as a **bar chart**. The problem becomes more complicated in the case of continuous numerical data. Here, the division (grouping) of the numerical data in certain classes (usually intervals) is necessary to draw the corresponding histogram. Concretely, to each lass (group) represented on one axis, there will correspond the relative frequency of occurrence (or the number of observations) on the other axis.

Experiment No. 2b

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Aim: Tutorial on Solving exercises in Data Preprocessing

Objective:
After completing this experiment, you will be able to understand the need of Data Preprocessing.

LO's to be achieved:

LON	Lab Outcomes
LO2	Identify sources of Data for mining and perform data exploration

Data Preprocessing:
Today's real-world databases are highly susceptible to noisy, missing, and inconsistent data due to their typically huge size (often several gigabytes or more) and their likely origin from multiple, heterogeneous sources. Low-quality data will lead o low-quality mining results.
There are several data preprocessing techniques. *Data cleaning* can be applied to remove noise and correct inconsistencies in data. *Data integration* merges data from multiple sources into a coherent data store such as a data warehouse. *Data reduction* can reduce data size by, for instance, aggregating, eliminating redundant features, or clustering. *Data transformations* (e.g., normalization) may be applied, where data are scaled to fall within a smaller range like 0.0 to 1.0. This can improve the accuracy and efficiency of mining algorithms involving distance measurements. These techniques are not mutually exclusive; they may work together. For example, data cleaning can involve transformations to correct wrong data, such as by transforming all entries for a *date* field to a common format.

Tasks in data preprocessing:

- Data cleaning: fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies.
- Data integration: using multiple databases, data cubes, or files.
- Data transformation: normalization and aggregation.
- Data reduction: reducng the volume but producing the same or similar analytical results.
- Data discretization: part of data reduction, replacing numerical attributes with nominal ones.

Data cleaning

1. Fill in missing values (attribute or class value):
    - Ignore the tuple: usually done when class label is missing.
    - Use the attribute mean (or majority nominal value) to fill in the missing value.
    - Use the attribute mean (or majority nominal value) for all samples belonging to the
      same class.
    - Predict the missing value by using a learning algorithm: consider the attribute with the missing value as a dependent (class) variable and run a learning algorithm (usually Bayes or decision tree) to predict the missing value.
2. Identify outliers and smooth out noisy data:
    - Binning
        - Sort the attribute values and partition them into bins (see "Unsupervised discretization" below);
        - Then smooth by bin means, bin median, or bin boundaries.
    - Clustering: group values in clusters and then detect and remove outliers (automatic or manual)
    - Regression: smooth by fitting the data into regression functions.

Correct inconsistent data: use domain knowledge or expert decision.

Data transformation

1. Normalization:
    - Scaling attribute values to fall within a specified range.
    - Example: to transform V in [min, max] to V' in [0,1], apply V'=(V-Min)/(Max-Min)
    - Scaling by using mean and standard deviation (useful when min and max are unknown or when there are outliers): V'=(V-Mean)/StDev
2. Aggregation: moving up in the concept hierarchy on numeric attributes.
3. Generalization: moving up in the concept hierarchy on nominal attributes.
4. Attribute construction: replacing or adding new attributes inferred by existing attributes.

Data reduction

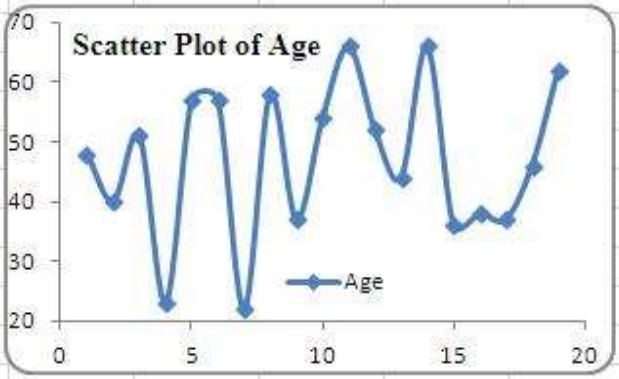
1. Reducing the number of attributes
    - Data cube aggregation: applying roll-up, slice or dice operations.
    - Removing irrelevant attributes: attribute selection (filtering and wrapper methods),
      Searching the attribute space (see Lecture 5: Attribute-oriented analysis).
    - Principle component analysis (numeric attributes only): searching for a lower dimensional space that can best represent the data..
2. Reducing the number of attribute values
    - Binning (histograms): reducing the number of attributes by grouping them into intervals (bins).
    - Clustering: grouping values in clusters.
    - Aggregation or generalization
3. Reducing the number of tuples
    - Sampling

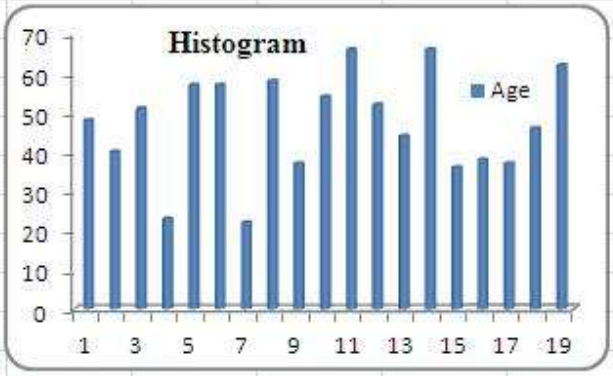
Discretization and generating concept hierarchies

1. Unsupervised discretization - class variable is not used.
    - Equal-interval (equiwidth) binning: split the whole range of numbers in intervals with equal size.
    - Equal-frequency (equidepth) binning: use intervals containing equal number of values.
2. Supervised discretization - uses the values of the class variable.
    - Using class boundaries. Three steps:
        - Sort values.
        - Place breakpoints between values belonging to different classes.
3. Generating concept hierarchies: recursively applying partitioning or discretization methods.

Data Exploration:

	A	B	C	D	E
1	Age	Age	Age	Age	Age
2	48	22	48	48	48
3	40	23	40	40	40
4	51	36	51	51	51
5	23	37	23	23	23
6	57	37	57	57	57
7	57	38	57	57	57
8	22	40	22	22	22
9	58	44	58	58	58
10	37	46	37	37	37
11	54	48	54	54	54
12	66	51	66	66	66
13	52	52	52	52	52
14	44	54	44	44	44
15	66	57	66	66	66
16	36	57	36	36	36
17	38	58	38	38	38
18	37	62	37	37	37
19	46	66	46	46	46
20	62	66	62	62	62
21	Mean & Av	Median	Mode	Variance	STD-DEV
22	47.05263	48	57	167.8304	12.95494





$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

Conclusion:

Basic statistical descriptions provide the analytical foundation for data preprocessing. The basic statistical measures for data summarization include mean, median, and mode for measuring the central tendency of data; and range, quantiles, quartiles, interquartile range, variance, and standard deviation for measuring the dispersion of data. Graphical representations (e.g., boxplots, histograms, and scatter plots) facilitate visual inspection of the data and are thus useful for data preprocessing and mining.

Data cleaning routines attempt to fill in missing values, smooth out noise while identifying outliers, and correct inconsistencies in the data. Data integration combines data from multiple sources to form a coherent data store. The resolution of semantic heterogeneity, metadata, correlation analysis, tuple duplication detection, and data conflict detection contribute to smooth data integration. Data reduction techniques obtain a reduced representation of the data while minimizing the loss of information content. These include methods of dimensionality reduction, numerosity reduction, and data compression. Data transformation routines convert the data into appropriate forms for mining.
Data discretization transforms numeric data by mapping values to interval or concept labels. Such methods can be used to automatically generate *concept hierarchies*for the data, which allows for mining at multiple levels of granularity. Discretization techniques include binning, histogram analysis, cluster analysis, decision tree analysis, and correlation analysis. For nominal data, concept hierarchies may be generated based on schema definitions as well as the number of distinct values per attribute.

Post Lab Exercise:

Qu-1: Suppose that the data for analysis includes the attribute *age*. The *age* values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

(a) What is the *mean* of the data? What is the *median*?
(b) What is the *mode* of the data? Comment on the data's modality (i.e., bimodal, trimodal, etc.).
(c) What is the *midrange* of the data?
(d) Can you find (roughly) the first quartile ($Q1$) and the third quartile ($Q3$) of the data?
(e) Give the *five-number summary* of the data.
(f) Show a *boxplot* of the data.
(g) How is a *quantile-quantile plot* different from a *quantile plot*?

Answer:
(a) What is the *mean* of the data? What is the *median*?

The (arithmetic) mean of the data is: $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = 809/27 = 30.$ The median (middle value
of the ordered set, as the number of values in the set is odd) of the data is: 25.

(b) What is the *mode* of the data? Comment on the data's modality (i.e., bimodal, trimodal, etc.). This data set has two values that occur with the same highest frequency and is, therefore, bimodal. The modes (values occurring with the greatest frequency) of the data are 25 and 35.

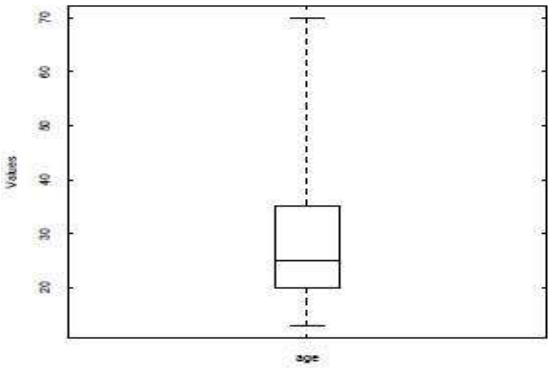
(c) What is the *midrange* of the data?
The midrange (average of the largest and smallest values in the data set) of the data is:

$(70 + 13)/2 = 41.5$

(d) Can you find (roughly) the first quartile ($Q1$) and the third quartile ($Q3$) of the data?
The first quartile (corresponding to the 25th percentile) of the data is: 20. The third quartile (corresponding to the 75th percentile) of the data is: 35.

(e) Give the *five-number summary* of the data.
The five number summary of a distribution consists of the minimum value, first quartile, median value, third quartile, and maximum value. It provides a good summary of the shape of the distribution and for this data is: 13, 20, 25, 35, 70.

(f) Show a *boxplot* of the data.



(g) How is a *quantile-quantile plot* different from a *quantile plot*?
A quantile plot is a graphical method used to show the approximate percentage of values below or equal to the independent variable in a univariate distribution. Thus, it displays quantile information for all the data, where the values measured for the independent variable are plotted against their corresponding quantile. A quantile-quantile plot however, graphs the quantiles of one univariate distribution against the corresponding quantiles of another univariate distribution. Both axes display the range of values measured for their corresponding distribution, and points are plotted that correspond to the quantile values of the two distributions. A line ($y = x$) can be added to the graph along with points representing where the first, second and third quantiles lie, in order to increase the graph's informational value. Points that lie above such a line indicate a correspondingly higher value for the distribution plotted on the y-axis, than for the distribution plotted on the x-axis at the same quantile. The opposite effect is true for points lying below this line.